

Support Surfaces

Support surfaces can be used on top of or instead of standard mattresses. They distribute pressure over a large surface area, and the type of support surface used will depend on the patient's needs and abilities.

Static support surfaces are recommended for patients who can assume several positions without bearing weight on an ulcer. These include foams, fluid- or air-filled mattresses, and overlays.

- **Foam** is probably the most commonly used material for pressure reduction, because it is easy to use, lightweight, and inexpensive. An investigative study showed that the combination of turning every 4 hours and the use of a viscoelastic polyurethane foam mattress decreased the incidence of pressure ulcers (grade II to IV) in at-risk patients from 14.3 percent to 3.0 percent.
- **Fluid-filled products** consist of small or large chambers filled with air, water, or other viscous fluid materials (such as silicone or polyvinyl). Most of them permit a high degree of immersion, allowing the body to sink into the surface as it conforms to bony prominences. This increases the surface pressure distribution area and lowers the interface pressure by transferring pressure to adjacent areas.

Dynamic support surfaces are indicated for patients who cannot tolerate frequent turning due to pain or severe illness, immobile patients, those with large or multiple ulcers, and patients with unresponsive ulcers. These provide cyclical, alternating pressure relief and include air-fluidized, low-air-loss, and alternating-air beds.

- **Air-fluidized beds** contain microspherical silicon-coated beads encased in an air-permeable fabric (polyester or Gore-Tex). The beads are suspended by pressurized streams of warmed air, which allows patients to "float" and decreases pressure through the principle of immersion while simultaneously reducing shear. Feces and body fluids flow freely through the sheet, keeping the skin dry.

- **Low-air-loss systems** use a series of connected, air-filled cushions inflated to specific pressures. Some have alternating and pulsating pressure features. A prospective, randomized trial demonstrated that the use of low-air-loss beds resulted in threefold faster wound contraction compared to foam mattresses. Unlike air-fluidized beds, urine and feces do not pass through the fabric of low-air-loss beds.
- **Alternating-pressure systems** distribute pressure by shifting the body weight to different surface contact areas. Air is pumped into chambers at periodic intervals to inflate and deflate them in opposite phases, thereby changing the location of contact pressure.

Wound Management

Cleansing

Cleansing should be performed gently to minimize mechanical and chemical trauma to healing tissues. **Normal saline** is preferred because it is most physiologic in relation to the wound. Safe and effective irrigation pressures for ulcers range from **4 to 15 psi**; higher pressures may cause tissue trauma and drive bacteria into wound tissue.

The following agents should be avoided because they are cytotoxic and may impair epithelialization: - Povidone-iodine - Sodium hypochlorite solution - Hydrogen peroxide - Acetic acid - Liquid detergents

Debridement

Ulcers with devitalized, necrotic tissue should be debrided, as necrotic tissue supports the growth of pathogenic organisms.

Mechanical debridement techniques include: - Moistened saline or wet-to-moist dressings - Hydrotherapy (whirlpool) - Wound irrigation - Dextranomer applications

Wet-to-dry dressings changed every 4 to 6 hours are painful and non-selective, removing both necrotic and healthy tissue. Whirlpool therapy may be considered for pressure ulcers with thick exudate, slough, or necrotic tissue. For wound irrigation, a **35-mL syringe** and **19-gauge angiographic catheter** can generate adequate pressure.

Sharp debridement involves the use of a scalpel, scissors, or other sharp instruments to remove devitalized tissue and is generally indicated when eschar is present.